

$$4.1) \quad z = ax + by + c$$

$$\sum_i (z_i - (ax_i + by_i + c))^2 \text{ minimalní}$$

$$(x, y, z) : (0, 0, 1) \rightarrow 1 = c$$

$$(1, 0, 1) \rightarrow 1 = a + c$$

$$(0, 1, 1) \rightarrow 1 = b + c$$

$$(1, 1, 2) \rightarrow 2 = a + b + c$$

$$(1, -1, 0) \rightarrow 0 = a - b + c$$

$$(-1, 1, 1) \rightarrow 1 = -a + b + c$$

$$\begin{array}{l} c = 1 \\ b + c = 1 \\ a - b + c = 0 \\ a + c = 1 \\ a + b + c = 2 \\ -a + b + c = 1 \end{array} \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & -1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{array} \right) \rightarrow \text{soustava nemá řešení}$$

Podle tvrzení 8.91 je množina přibližných řešení soustavy $A\vec{x} = \vec{b}$ metodou nejmenších čtverců rovna množině všech průměrných řešení soustavy

$$A^T A \hat{\vec{x}} = A^T \vec{b} \quad \hat{\vec{x}} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \begin{array}{c} A \\ \parallel \\ \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & -1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{array} \right) \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & -1 \\ 1 & 1 & -1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \quad \left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ 0 & 15 & 10 & 18 \\ 0 & 5 & 10 & 10 \end{array} \right) \sim \left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ 0 & 15 & 10 & 18 \\ 0 & 0 & 20 & 12 \end{array} \right)$$

$$20c = 12 \quad c = \frac{3}{5} \quad 15b + 10c = 18 \quad 15b = 18 - 6 \quad b = \frac{4}{5}$$

$$4a - b + 2c = 2 \quad 4a = 2 + \frac{4}{5} - \frac{6}{5} \quad 4a = \frac{8}{5} \quad a = \frac{2}{5}$$

$$a = \frac{2}{5} \quad b = \frac{4}{5} \quad c = \frac{3}{5}$$

$$\text{Rovnice rovin: } \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5} = z$$

$$\sum_i (z_i - (ax_i + by_i + c))^2 = \frac{4}{25} + \frac{4}{25} + \frac{1}{25} + \frac{1}{25} = \frac{10}{25}$$

$$(0, 0, 1) \rightarrow \left(1 - \frac{3}{5}\right)^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

$$(1, 0, 1) \rightarrow \left(1 - \frac{2}{5} - \frac{3}{5}\right)^2 = 0^2 = 0$$

$$(0, 1, 1) \rightarrow \left(1 - \frac{4}{5} - \frac{3}{5}\right)^2 = \left(-\frac{2}{5}\right)^2 = \frac{4}{25}$$

$$(1, 1, 2) \rightarrow \left(2 - \frac{2}{5} - \frac{4}{5} - \frac{3}{5}\right)^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

$$(1, -1, 0) \rightarrow \left(0 - \frac{2}{5} + \frac{4}{5} - \frac{3}{5}\right)^2 = \left(-\frac{1}{5}\right)^2 = \frac{1}{25}$$

$$(-1, 1, 1) \rightarrow \left(1 + \frac{2}{5} - \frac{4}{5} - \frac{3}{5}\right)^2 = 0$$